

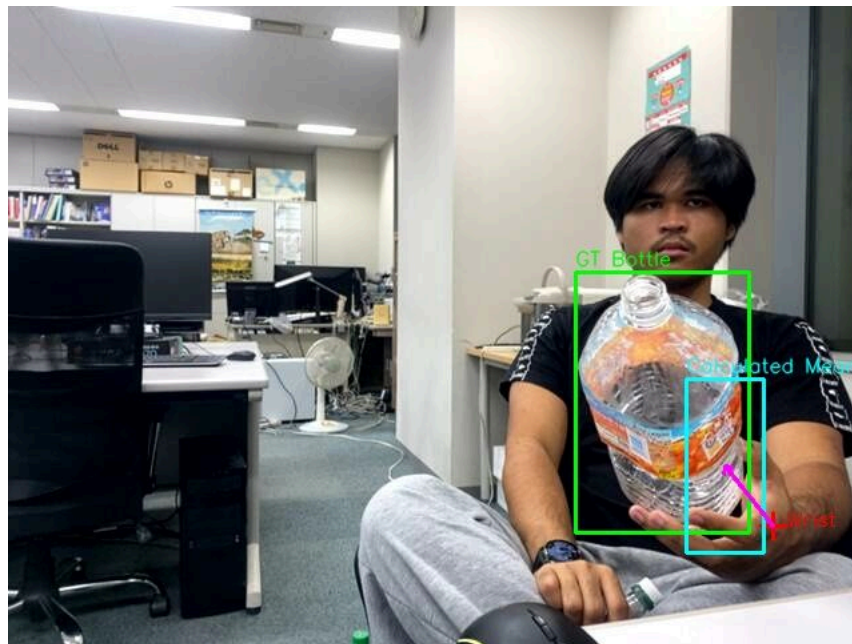
10th Meeting 1/13

What I said will deliver in 10th meeting:

- Full story of my architecture model and all of the methods that used for this training

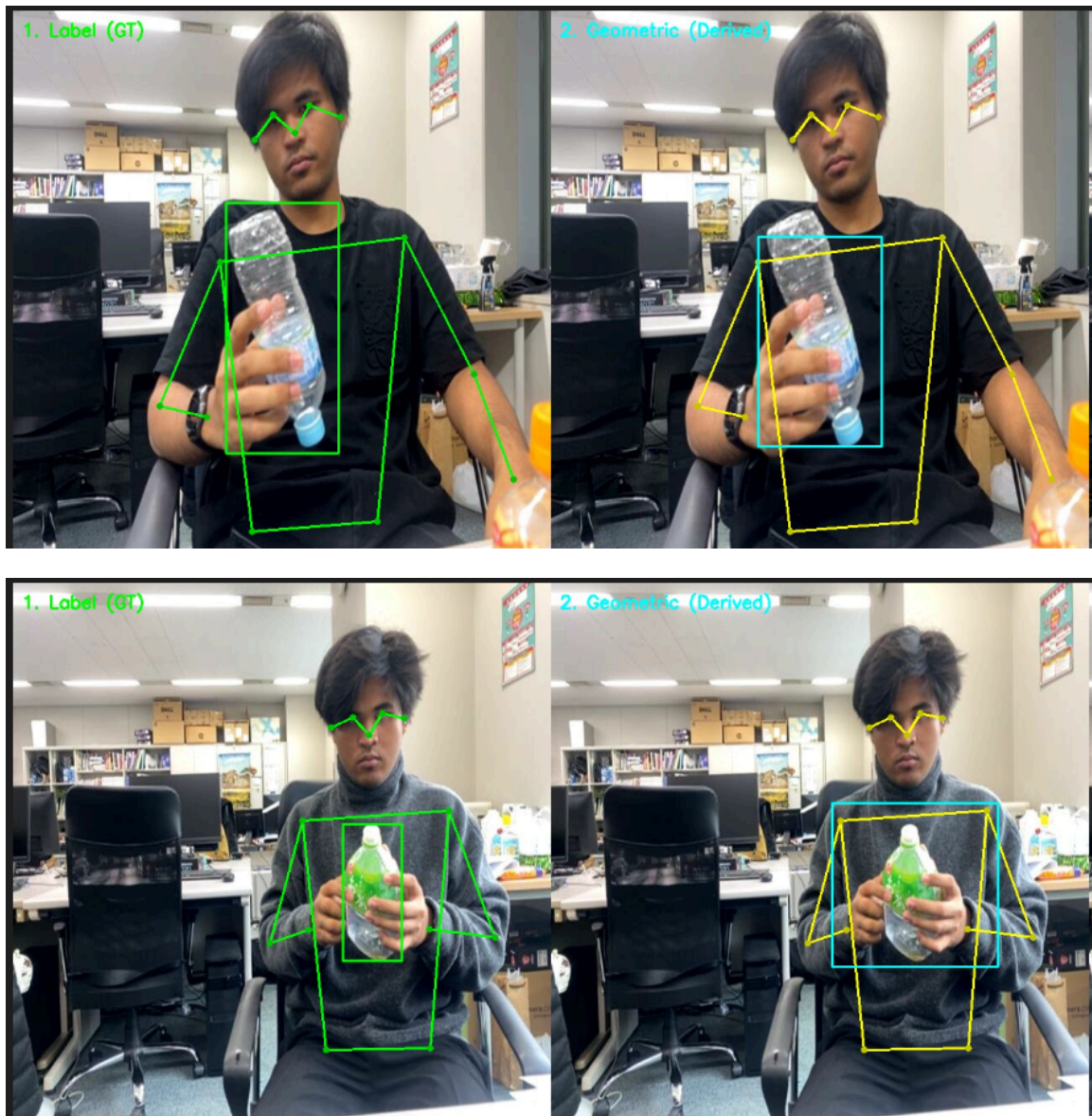
I am currently in the model methodology exploration and still finding the best methodology or the best way to train my model

Previously, I was having problem using geometric



The update:

- I already make it more standardize and match into every dataset



From the previous meeting, you were advising to use the angle of the body etc. But after evaluating, there is not a lot correlation between the shoulders angle into the holding position

- The data result of the experiments

The model is still on training. I encounter many errors for the implementation previously.

The idea is: limit the detection head near to the heatmap generated by geometric in P3 P4 and P5 (not only RGB but heatmap or cut the space)

40/300 epochs



I am currently train to compare the proposed solution (using keypoints to guide detection by geometric), also the base model with several conditions. But I am still not confident about the result and the training file. I will share it next week

- comparative study compare to common dataset

[https://www.google.com/url?](https://www.google.com/url?sa=t&source=web&rct=j&opi=89978449&url=https://ikeaasm.github.io/&ved=2ahUKewiL4ob_IleSAxU7kK8BHfK2NJYQFnoE)

[sa=t&source=web&rct=j&opi=89978449&url=https://ikeaasm.github.io/&ved=2ahUKewiL4ob_IleSAxU7kK8BHfK2NJYQFnoE](https://ikeaasm.github.io/&ved=2ahUKewiL4ob_IleSAxU7kK8BHfK2NJYQFnoE)
- IKEA ASM Dataset

<https://arxiv.org/pdf/2007.00394>

https://openaccess.thecvf.com/content/WACV2024/papers/Ben-Shabat_IKEA_Ego_3D_Dataset_Understanding_Furniture_Assembly_Actions_From_Ego-View_WACV_2024_paper.pdf

40Gb data → IKEA

Number of the data →

30GB data HA4M

<https://www.nature.com/articles/s41597-022-01843-z>

https://opus.lib.uts.edu.au/bitstream/10453/172613/2/Assembly_Action_Recognition_in_Smart_Manufacturing.pdf

[https://pdf.sciencedirectassets.com/280203/1-s2.0-S1877050924X00162/1-s2.0-S187705092402489X/main.pdf?X-Amz-Security-](https://pdf.sciencedirectassets.com/280203/1-s2.0-S1877050924X00162/1-s2.0-S187705092402489X/main.pdf?X-Amz-Security-Token=IQoJb3JpZ2luX2VjEC8aCXVzLWVhc3QtMSJHMEUCIcmzcUppqjsxobQHdmt3uzZKOUo1UVX%2Fty9QgQZmD0ZbCAiAmz-Algorithm=AWS4-HMAC-SHA256&X-Amz-Date=20260112T235215Z&X-Amz-SignedHeaders=host&X-Amz-Expires=300&X-Amz-Credential=ASIAQ3PHCVTYXLCIOXNE%2F20260112%2Fus-east-1%2Fs3%2Faws4_request&X-Amz-Signature=72f53e12d20e68f895e0394a203110aa3f9bb6ff82cb94f61a0e7a2fcc10996&hash=65f91fd7315e07be58b0ca421df48e49-ddd2-4f1e-988a-a2d6ca310744&sid=db2b3844169d454f2b7b8597bc2adae9025fgxrqa&type=client&tsoh=d3d3LnNjaWVuY2VkaXJlY3QuY2)

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<https://ceur-ws.org/Vol-3762/505.pdf>

- The complete end-to-end prototype for this model (hopefully)

I am not sure if I can finish the end to end prototype with these several experiments. I will focus on achieving my model first and try it to public dataset. After it is validated, then I will continue to the end to end prototype. This might affect the postponed conference schedule.

The goal next week:

- Show the result of ablation training (comparison from my model to the different model including the base model)
- Explain the full architecture and loss function of my model

For the next 2 weeks:

- Try the public dataset

Next 3 weeks:

- Evaluate the result of the public dataset → if good: implement the full prototype